

Enabling Comprehensive Data Monitoring for Solar Farms Nationwide

A Case Study for the Cloud Service Toolkit

Background

As solar farms and power plants began to multiply across the country, so did the challenges in monitoring site conditions and power production data. To keep a solar farm running at peak operating performance, you need to know exactly what's going on at the site. This means having real-time access to a mountain of data that is collected from a variety of sensors and devices, measuring such things as temperature, power output, voltage, solar irradiance, data from weather stations, and so on.

Panasonic Eco Solutions quickly realized that being able to monitor the data from any number of installations and make that data readily available through an easy-to-use dashboard would provide a huge benefit to the solar farms, both individually and collectively. But to do this, Panasonic knew they had to avoid vendor lock-in, which meant that the system would need to integrate with many existing protocols and systems without having to rework the back-end systems for each one. The system would also need to accommodate new protocols quickly and seamlessly.

6

Current Energy Providers

126

Current Solar Installations

28,000

Data Points per Hour

20,000,000

Data Points per Month

Challenges

The problem seemed overwhelming. How do you gather the data from 126 different solar installations spread out from California to New Jersey and present it in a dashboard interface?

The project presented a number of challenges:

1. **Scale.** The system would have to be able to handle over a hundred solar installations initially and scale easily as more solar farms were added.
2. **Integrating with different vendors.** Analyzing data would be problematic because each device and sensor produced data in a different format and using different protocols.
3. **Data storage and Dashboard displays.** The data would need to be available for realtime monitoring as well as for analysis.
4. **Security.** The data would need to be secure against unauthorized access or use.

The Solution

Panasonic Salt Lake City Laboratory (PSLCL) found the solution using the Cloud Service Toolkit (CST) and OpenDOF (www.opendof.org), a freely available open source project. OpenDOF allowed them to define a common data model that incorporated security. Using the OpenDOF interface repository (<https://interface.opendof.org>), interface definitions were developed, approved, and shared. An OpenDOF cloud service was created to bridge data from the various vendors through the CST, so data can be fed to dashboards and analytics processing in a normalized, consistent, and secure way.



Results

Using the system created through OpenDOF and CST, every piece of equipment at each of the solar sites (sensors, solar panels, inverters, meters, weather stations) is now securely monitored, 24/7 365 days a year. What's more, the collected data is monitored in Panasonic's operations center on a dashboard and is used to determine when to schedule maintenance visits and to discover trends.

Integrating new equipment is not a problem. An OpenDOF cloud service was created to bridge data from the various vendors through the CST. This cloud service was built using a plug-in architecture that makes it easy to incorporate new vendors and equipment. When a new solar data vendor or piece of equipment needs to be added to the system, a new plug-in adapter can be created quickly—typically in less than two weeks—and brought online without interruption or downtime. (To date, two new adapters have been added with no impact on the rest of the system—databases, dashboards, web applications, alarming, analytics, etc.) Data is never tied to a specific vendor, ensuring there is no vendor lock-in. What's more, the CST is built on OpenDOF, which is a freely available, open-source project that is backed by a patent pledge from Panasonic and built on more than a decade of experience in device networking.

Currently, the CST services the bridge protocols and provides a unified data model for 6 energy providers across 126 solar installations with many more planned, collecting over 28,000 data points per hour and over 20 million data points per month.



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Real Solutions Today

Thanks to OpenDOF and the CST, Panasonic is saving time and money by avoiding frequent updates to the monitoring dashboard and analytics engine, while still

supporting a wide variety of solar vendors and equipment in the field. Having one consistent view of the data means operations are simpler and easier to manage across the entire set of solar farms. What had once seemed an overwhelming and daunting task is now a manageable, consistent, and secure solution.

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